

Laboration 1 8F1693

1. $u(x_1, 1) = g(x_1)$, $\Omega := \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \in \mathbb{R}, x_2 > 1\}$

$$\bar{V}(x_1, x_2) = (x_1, x_2),$$

$$f(x_1, x_2) = 2u(x_1, x_2), \quad g: \mathbb{R} \rightarrow \mathbb{R} \text{ g\u00e4ven.}$$

L\u00f6s analytiskt $\bar{V}(\bar{x}) \cdot \nabla u(\bar{x}) = 2u(\bar{x})$.

L\u00f6sn. L\u00f6s $\bar{x}(s) = \begin{pmatrix} x_1(s) \\ x_2(s) \end{pmatrix}$. Vara karakteristikan till v.

D\u00e5 $\bar{x}' = \bar{V}(\bar{x}(s)) \longrightarrow \begin{cases} x_1' = x_1(s) \\ x_2' = x_2(s) \end{cases}$ Ans: $\longrightarrow$

$$\begin{cases} x_1(s) = c_1 e^s \\ x_2(s) = c_2 e^s \end{cases}$$

L\u00f6s nu $\bar{x}(0) = (e, 1) \stackrel{(*)}{\Rightarrow} \begin{cases} x_1(s) = e e^s \\ x_2(s) = e^s \end{cases}$, e parameter $\in \mathbb{R}$.

D\u00e5 $u(\bar{x}(0)) = u(e, 1) = g(e)$.

Nu g\u00f6r vi $\frac{d}{ds} u(\bar{x}(s)) = \bar{x}'(s) \cdot \nabla u(\bar{x}(s)) = 2u(\bar{x}(s))$

S\u00e5 g\u00f6r ODE:n $\left\{ \begin{array}{l} \text{skriv } u(\bar{x}(s)) = u(s) \\ u'(s) = 2u(s) \end{array} \right\} \Rightarrow u(s) = c e^{2s}$. Med $u(0) = g(e) \Rightarrow u(s) = g(e) e^{2s}$.

ODE-systemet $(*)$ g\u00e4r nu $s = \ln(x_1/e)$, $s = \ln(x_2)$

& vi f\u00f6r $x_2 = x_1/e$, $e = x_1/x_2$. Insett i $u(s)$ g\u00e4r

$$u(x_1, x_2) = g\left(\frac{x_1}{x_2}\right) e^{\ln x_2}.$$